

Q2

The screenshot shows the MongoDB Atlas web interface. The browser address bar displays the URL: `cloud.mongodb.com/v2/6267e61fc1641a0c957b7492#/clusters`. The page title is "Clusters". On the left, a sidebar menu lists various database management tools. The main content area shows details for "Cluster0", including a "Visualize Your Data" section with charts for Read/Write operations, Connections, In/Out traffic, and Data Size. Below this, a table lists cluster specifications.

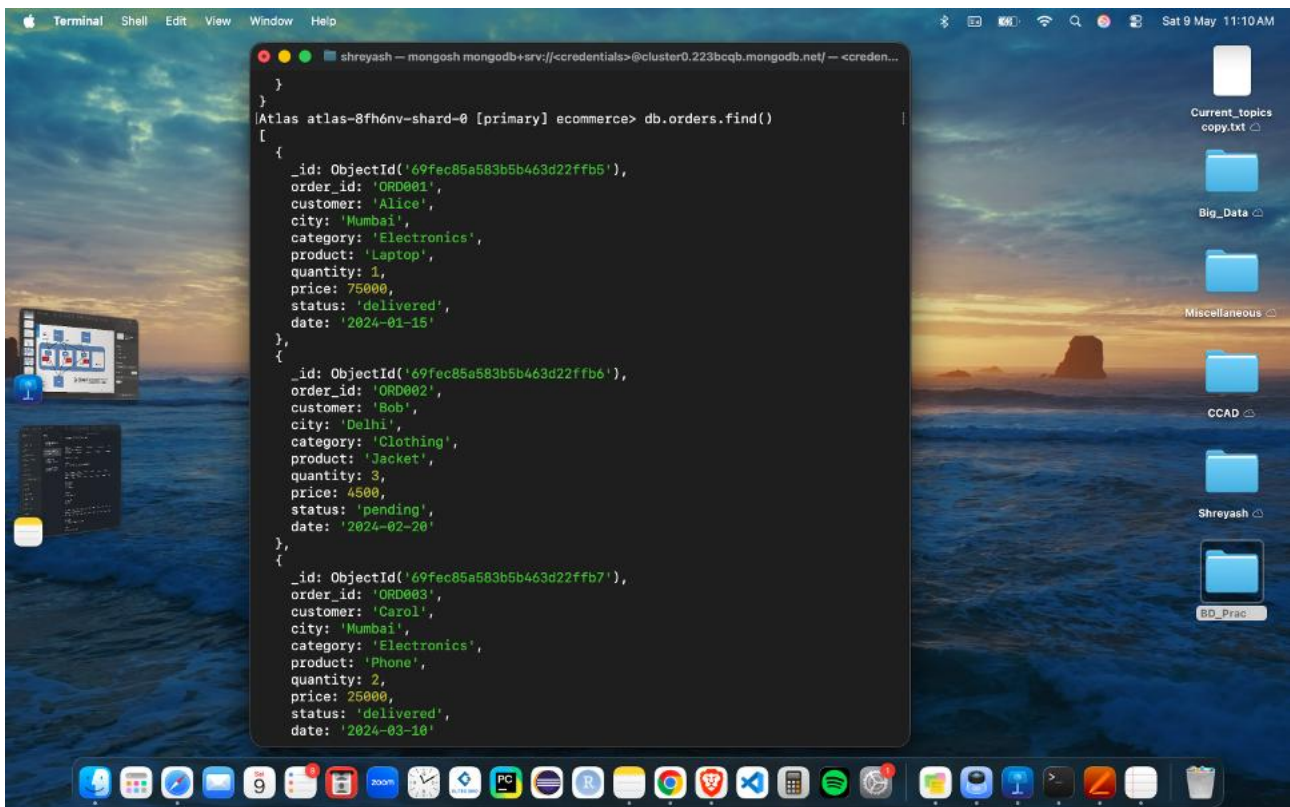
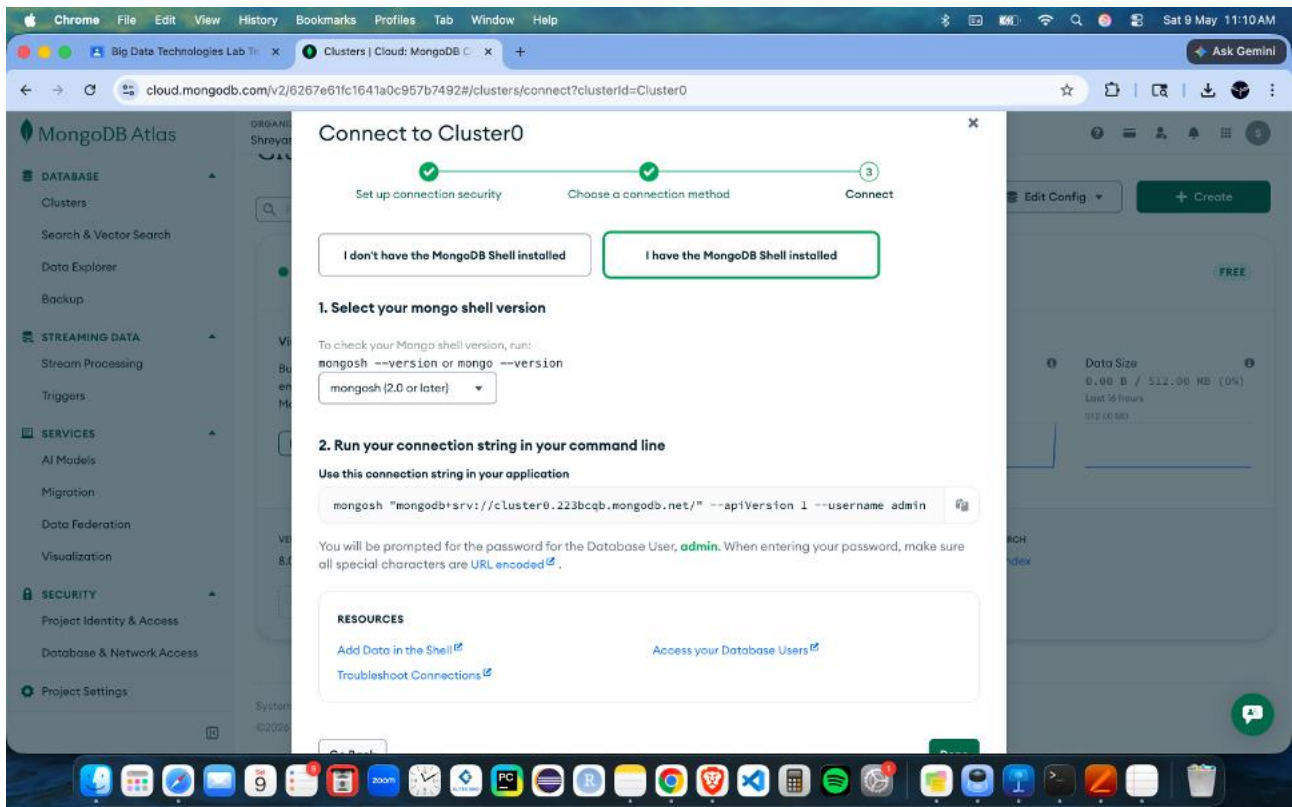
VERSION	REGION	TYPE	BACKUPS	LINKED APP SERVICES	ATLAS SQL	ATLAS SEARCH
8.0.23	AWS / Mumbai (ap-south-1)	Replica Set - 3 nodes	Inactive	None Linked	Connect	Create Index

The screenshot shows a terminal window with the following commands and output:

```
shreyash@Shreyashs-MacBook-Pro ~ % mongosh "mongodb+srv://cluster0.223bcqb.mongodb.net/" --apiVersion 1 --username admin
Enter password: ****
Current Mongosh Log ID: 69fec82d583b5b463d22ffb4
Connecting to: mongodb+srv://cluster0.223bcqb.mongodb.net/?appName=mongosh+2.8.2
Using MongoDB: 8.0.23 (API Version 1)
Using Mongosh: 2.8.2
mongosh 2.8.3 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/

Atlas atlas-8fh6nv-shard-0 [primary] test> use ecommerce
switched to db ecommerce
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce> db.orders.insertMany([ { order_id: "ORD001", customer: "Alice", city: "Mumbai", category: "Electronics", product: "Laptop", quantity: 1, price: 75000, status: "delivered", date: "2024-01-15" }, { order_id: "ORD002", customer: "Bob", city: "Delhi", category: "Clothing", product: "Jacket", quantity: 3, price: 4500, status: "pending", date: "2024-02-20" }, { order_id: "ORD003", customer: "Carol", city: "Mumbai", category: "Electronics", product: "Phone", quantity: 2, price: 25000, status: "delivered", date: "2024-03-10" }, { order_id: "ORD004", customer: "Dave", city: "Pune", category: "Books", product: "Big Data Guide", quantity: 5, price: 800, status: "shipped", date: "2024-03-25" }, { order_id: "ORD005", customer: "Eve", city: "Delhi", category: "Electronics", product: "Tablet", quantity: 1, price: 35000, status: "delivered", date: "2024-03-15" } ])
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce> db.orders.insertMany([ { order_id: "ORD001", customer: "Alice", city: "Mumbai", category: "Electronics", product: "Laptop", quantity: 1, price: 75000, status: "delivered", date: "2024-01-15" }, { order_id: "ORD002", customer: "Bob", city: "Delhi", category: "Clothing", product: "Jacket", quantity: 3, price: 4500, status: "pending", date: "2024-02-20" }, { order_id: "ORD003", customer: "Carol", city: "Mumbai", category: "Electronics", product: "Phone", quantity: 2, price: 25000, status: "delivered", date: "2024-03-10" }, { order_id: "ORD004", customer: "Dave", city: "Pune", category: "Books", product: "Big Data Guide", quantity: 5, price: 800, status: "shipped", date: "2024-03-25" }, { order_id: "ORD005", customer: "Eve", city: "Delhi", category: "Electronics", product: "Tablet", quantity: 1, price: 35000, status: "delivered", date: "2024-03-15" } ])
```



```
Terminal Shell Edit View Window Help Sat 9 May 11:13 AM
shreyash — mongosh mongodb+srv://<credentials>@cluster0.223bcqb.mongodb.net/ — <creden...
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce> db.orders.find({ status: "delivered" }).pretty()
[[{"_id": ObjectId("69fec85a583b5b463d22ffb5"),
  order_id: "ORD001",
  customer: "Alice",
  city: "Mumbai",
  category: "Electronics",
  product: "Laptop",
  quantity: 1,
  price: 75000,
  status: "delivered",
  date: "2024-01-15"},
{"_id": ObjectId("69fec85a583b5b463d22ffb7"),
  order_id: "ORD003",
  customer: "Carol",
  city: "Mumbai",
  category: "Electronics",
  product: "Phone",
  quantity: 2,
  price: 25000,
  status: "delivered",
  date: "2024-03-10"},
{"_id": ObjectId("69fec85a583b5b463d22ffb9"),
  order_id: "ORD005",
  customer: "Eve",
  city: "Delhi",
  category: "Electronics",
  product: "Tablet",
  quantity: 1,
  price: 35000,
  status: "delivered",
  date: "2024-03-10"}]]
```

```
Terminal Shell Edit View Window Help Sat 9 May 11:13 AM
shreyash — mongosh mongodb+srv://<credentials>@cluster0.223bcqb.mongodb.net/ — <creden...
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce> db.orders.updateOne(
  { order_id: "ORD002" },
  { $set: { status: "shipped" } }
)
acknowledged: true,
insertedId: null,
matchedCount: 1,
modifiedCount: 1,
upsertedCount: 0
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce>
```

```
Terminal Shell Edit View Window Help Sat 9 May 11:14 AM

shreyash — mongosh mongodb+srv://<credentials>@cluster0.223bcqb.mongodb.net/ — <creden...

order_id: 'ORD005',
customer: 'Eve',
city: 'Delhi',
category: 'Electronics',
product: 'Tablet',
quantity: 1,
price: 35000,
status: 'delivered',
date: '2024-04-05'
}
}
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce> db.orders.updateOne(
  { order_id: "ORD002" },
  { $set: { status: "shipped" } }
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce> db.orders.insertOne({
  order_id: "ORD006",
  customer: "Frank",
  city: "Bangalore",
  category: "Electronics",
  product: "Smart Watch",
  quantity: 2,
  price: 12000,
  status: "pending",
  date: "2024-05-10"
})
{
  acknowledged: true,
  insertedId: ObjectId('69fec9a2583b5b463d22ffba')
}
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce>
```

```
Terminal Shell Edit View Window Help Sat 9 May 11:14 AM

shreyash — mongosh mongodb+srv://<credentials>@cluster0.223bcqb.mongodb.net/ — <creden...

category: 'Electronics',
product: 'Tablet',
quantity: 1,
price: 35000,
status: 'delivered',
date: '2024-04-05'
}
}
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce> db.orders.updateOne(
  { order_id: "ORD002" },
  { $set: { status: "shipped" } }
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce> db.orders.insertOne({
  order_id: "ORD006",
  customer: "Frank",
  city: "Bangalore",
  category: "Electronics",
  product: "Smart Watch",
  quantity: 2,
  price: 12000,
  status: "pending",
  date: "2024-05-10"
})
{
  acknowledged: true,
  insertedId: ObjectId('69fec9a2583b5b463d22ffba')
}
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce> db.orders.deleteOne({ order_id: |
"ORD004" })
{ acknowledged: true, deletedCount: 1 }
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce>
```

```
Terminal Shell Edit View Window Help Sat 9 May 11:15 AM

shreyash — mongosh mongodb+srv://<credentials>@cluster0.223bcqb.mongodb.net/ — <creden...

Atlas atlas-8fh6nv-shard-0 [primary] ecommerce> db.orders.find().pretty()

[
  {
    _id: ObjectId('69fec85a583b5b463d22ffb5'),
    order_id: 'ORD001',
    customer: 'Alice',
    city: 'Mumbai',
    category: 'Electronics',
    product: 'Laptop',
    quantity: 1,
    price: 75000,
    status: 'delivered',
    date: '2024-01-15'
  },
  {
    _id: ObjectId('69fec85a583b5b463d22ffb6'),
    order_id: 'ORD002',
    customer: 'Bob',
    city: 'Delhi',
    category: 'Clothing',
    product: 'Jacket',
    quantity: 3,
    price: 4500,
    status: 'shipped',
    date: '2024-02-20'
  },
  {
    _id: ObjectId('69fec85a583b5b463d22ffb7'),
    order_id: 'ORD003',
    customer: 'Carol',
    city: 'Mumbai',
    category: 'Electronics',
    product: 'Phone',
    quantity: 2,
    price: 25000,
    status: 'delivered',
    date: '2024-03-10'
  },
]
```

```
Terminal Shell Edit View Window Help Sat 9 May 11:14 AM

shreyash — mongosh mongodb+srv://<credentials>@cluster0.223bcqb.mongodb.net/ — <creden...

},
{
  _id: ObjectId('69fec85a583b5b463d22ffb7'),
  order_id: 'ORD003',
  customer: 'Carol',
  city: 'Mumbai',
  category: 'Electronics',
  product: 'Phone',
  quantity: 2,
  price: 25000,
  status: 'delivered',
  date: '2024-03-10'
},
{
  _id: ObjectId('69fec85a583b5b463d22ffb9'),
  order_id: 'ORD005',
  customer: 'Eve',
  city: 'Delhi',
  category: 'Electronics',
  product: 'Tablet',
  quantity: 1,
  price: 35000,
  status: 'delivered',
  date: '2024-04-05'
},
{
  _id: ObjectId('69fec9a2583b5b463d22ffba'),
  order_id: 'ORD006',
  customer: 'Frank',
  city: 'Bangalore',
  category: 'Electronics',
  product: 'Smart Watch',
  quantity: 2,
  price: 12000,
  status: 'pending',
  date: '2024-05-10'
}
]
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce>
```

```
Terminal Shell Edit View Window Help Sat 9 May 11:16 AM

shreyash — mongosh mongodb+srv://<credentials>@cluster0.223bcqb.mongodb.net/ — <creden...

price: 35000,
status: 'delivered',
date: '2024-04-05'
},
{
  _id: ObjectId('69fec9a2583b5b463d22ffba'),
  order_id: 'ORD006',
  customer: 'Frank',
  city: 'Bangalore',
  category: 'Electronics',
  product: 'Smart Watch',
  quantity: 2,
  price: 12000,
  status: 'pending',
  date: '2024-05-10'
}
]
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce> db.orders.aggregate([
  {
    $group: {
      _id: "$category",
      total_revenue: {
        $sum: {
          $multiply: ["$price", "$quantity"]
        }
      }
    },
    $sort: {
      total_revenue: -1
    }
  }
])
[
  { _id: 'Electronics', total_revenue: 184000 },
  { _id: 'Clothing', total_revenue: 13500 }
]
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce>
```

```
Terminal Shell Edit View Window Help Sat 9 May 11:17 AM

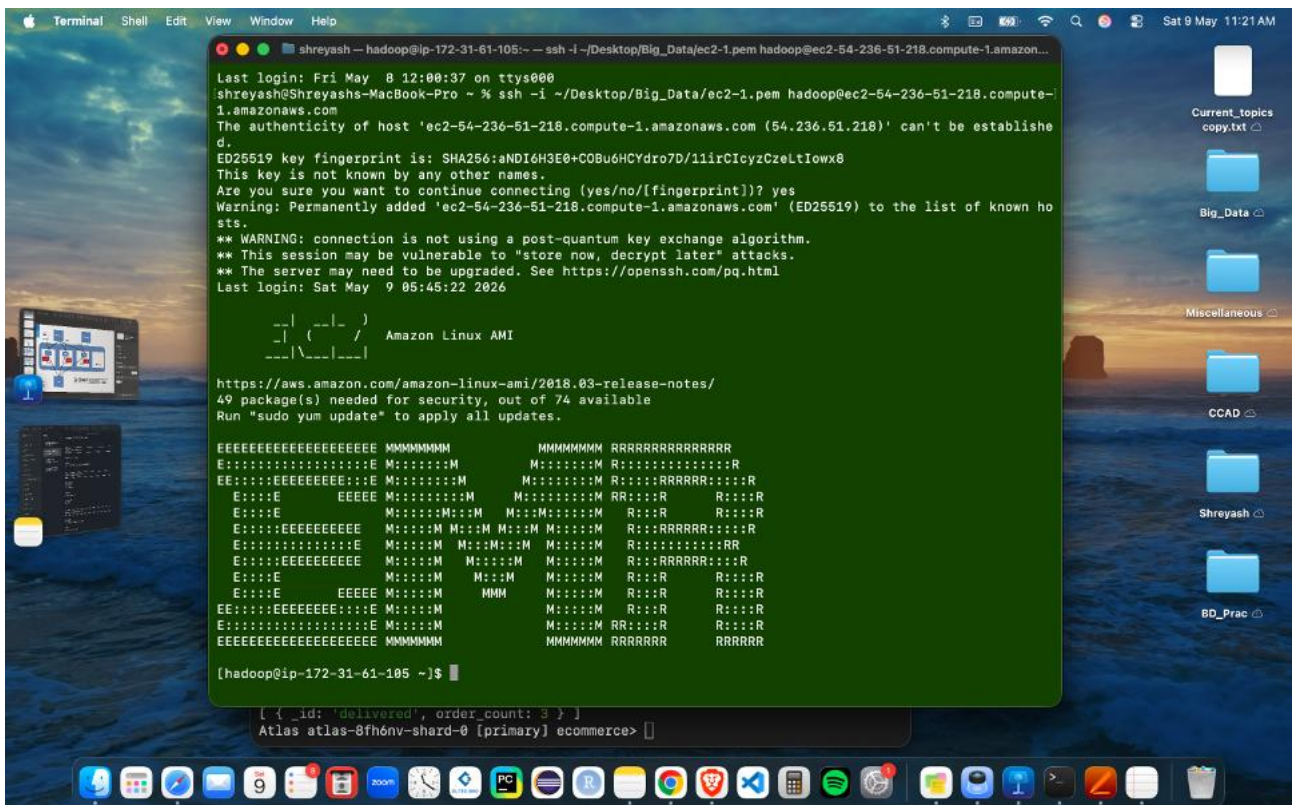
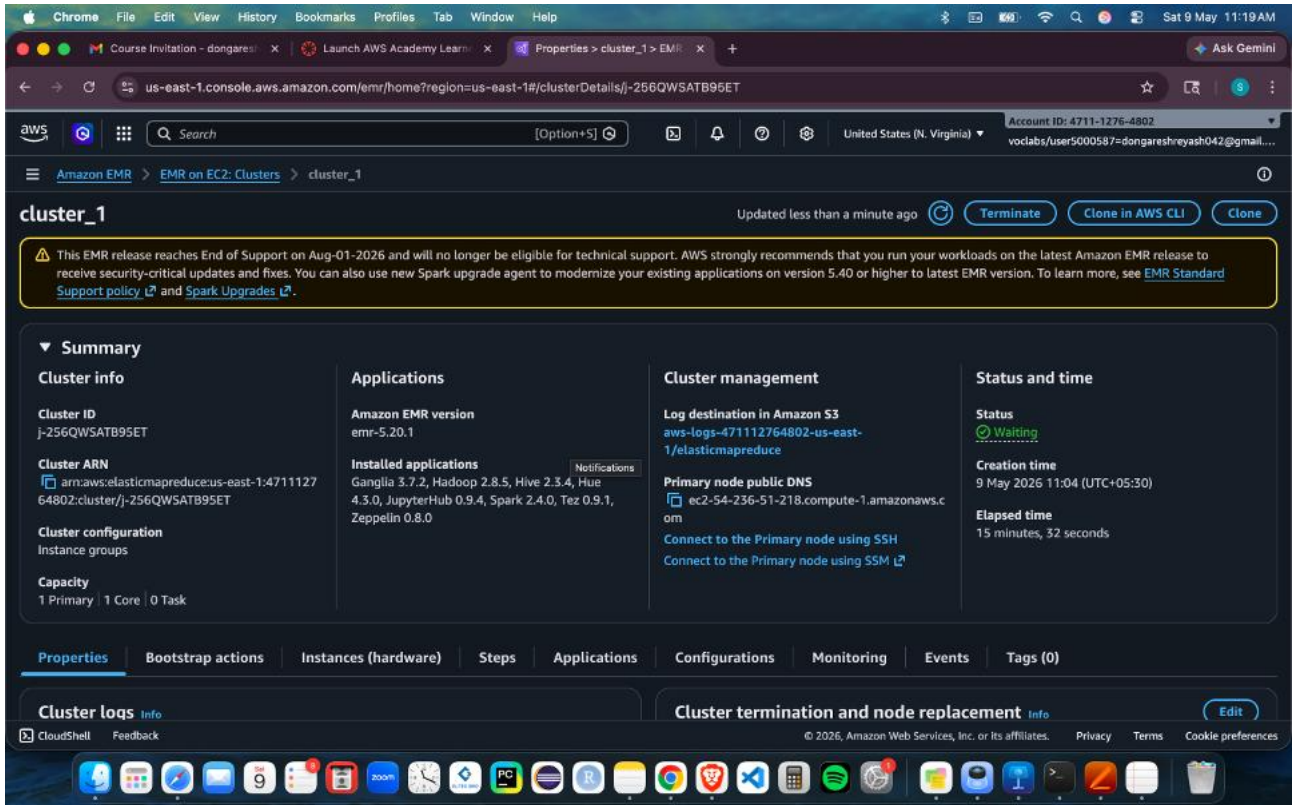
shreyash — mongosh mongodb+srv://<credentials>@cluster0.223bcqb.mongodb.net/ — <creden...

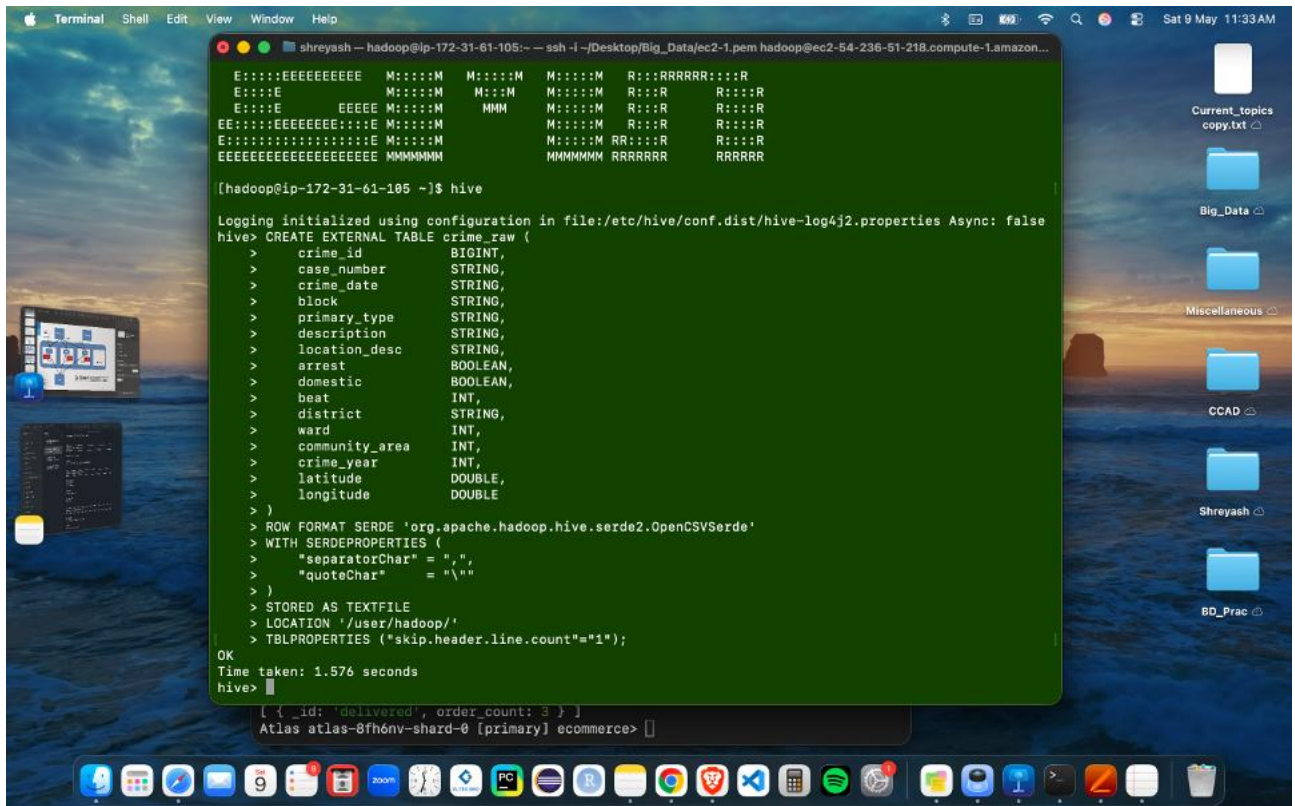
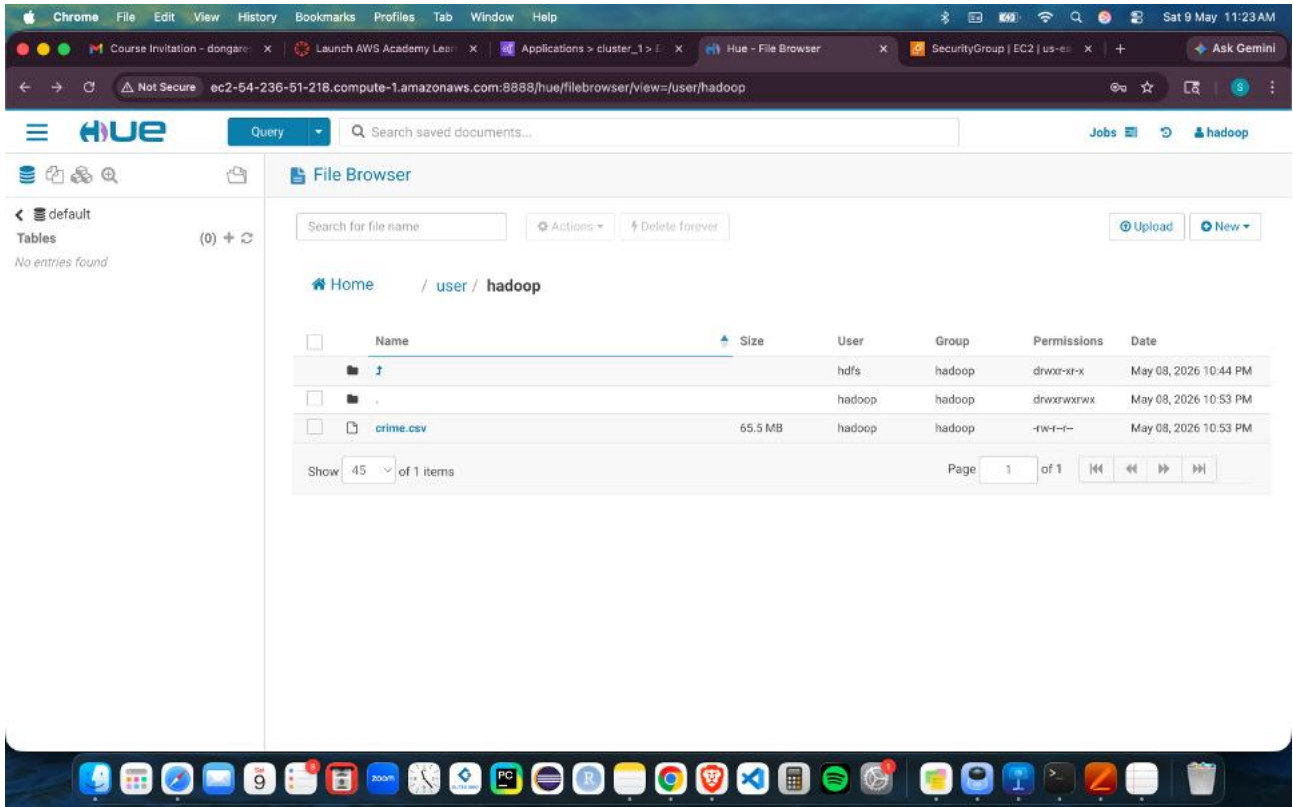
$group: {
  _id: "$category",
  total_revenue: {
    $sum: {
      $multiply: ["$price", "$quantity"]
    }
  }
},
{
  $sort: {
    total_revenue: -1
  }
}
])
[
  { _id: 'Electronics', total_revenue: 184000 },
  { _id: 'Clothing', total_revenue: 13500 }
]
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce> db.orders.aggregate([
  {
    $group: {
      _id: "$city",
      total_orders: { $sum: 1 },
      avg_price: { $avg: "$price" }
    },
    $sort: {
      total_orders: -1
    }
  }
])
[
  { _id: 'Delhi', total_orders: 2, avg_price: 19750 },
  { _id: 'Mumbai', total_orders: 2, avg_price: 50000 },
  { _id: 'Bangalore', total_orders: 1, avg_price: 12000 }
]
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce>
```

```
Terminal Shell Edit View Window Help Sat 9 May 11:19 AM
shreyash — mongosh mongodb+srv://<credentials>@cluster0.223bcqb.mongodb.net/ — <creden...

[ ]
[
  { _id: 'Electronics', total_revenue: 184000 },
  { _id: 'Clothing', total_revenue: 13500 }
]
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce> db.orders.aggregate([
  {
    $group: {
      _id: "$city",
      total_orders: { $sum: 1 },
      avg_price: { $avg: "$price" }
    },
    $sort: {
      total_orders: -1
    }
  }
])
[
  { _id: 'Delhi', total_orders: 2, avg_price: 19750 },
  { _id: 'Mumbai', total_orders: 2, avg_price: 50000 },
  { _id: 'Bangalore', total_orders: 1, avg_price: 12000 }
]
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce> db.orders.aggregate([
  {
    $group: {
      _id: "$status",
      order_count: { $sum: 1 }
    },
    $match: {
      order_count: { $gt: 1 }
    }
  }
])
[ { _id: 'delivered', order_count: 3 } ]
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce>
```

Q1





```
Terminal Shell Edit View Window Help Sat 9 May 11:34 AM
shreyash -- hadoop@ip-172-31-61-105:~ -- ssh -i ~/Desktop/Big_Data/ec2-1.pem hadoop@ec2-54-236-51-218.compute-1.amazonaws.com

> crime_year INT,
> latitude DOUBLE,
> longitude DOUBLE
>
> ROW FORMAT SERDE 'org.apache.hadoop.hive.serde2.OpenCSVSerde'
> WITH SERDEPROPERTIES (
>   "separatorChar" = ",",
>   "quoteChar" = "\""
> )
> STORED AS TEXTFILE
> LOCATION '/user/hadoop/'
> TBLPROPERTIES ("skip.header.line.count"="1");
OK
Time taken: 1.576 seconds
hive> CREATE TABLE crimes_partitioned (
>   crime_id BIGINT,
>   case_number STRING,
>   crime_date STRING,
>   block STRING,
>   primary_type STRING,
>   description STRING,
>   location_desc STRING,
>   arrest BOOLEAN,
>   domestic BOOLEAN,
>   beat INT,
>   district STRING,
>   ward INT,
>   community_area INT,
>   latitude DOUBLE,
>   longitude DOUBLE
> )
> PARTITIONED BY (crime_year INT)
> CLUSTERED BY (district) INTO 4 BUCKETS
> ROW FORMAT DELIMITED
> FIELDS TERMINATED BY ','
> STORED AS ORC;
OK
Time taken: 0.471 seconds
hive>

[ { _id: 'delivered', order_count: 3 } ]
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce>
```

```
Terminal Shell Edit View Window Help Sat 9 May 11:46 AM
shreyash -- hadoop@ip-172-31-61-105:~ -- ssh -i ~/Desktop/Big_Data/ec2-1.pem hadoop@ec2-54-236-51-218.compute-1.amazonaws.com

Time taken: 0.481 seconds, Fetched: 5 row(s)
hive>
> SET hive.exec.dynamic.partition = true;
hive> SET hive.exec.dynamic.partition.mode = nonstrict;
hive>
> INSERT OVERWRITE TABLE crimes_partitioned
> PARTITION (crime_year)
> SELECT
>   crime_id,
>   case_number,
>   crime_date,
>   block,
>   primary_type,
>   description,
>   location_desc,
>   arrest,
>   domestic,
>   beat,
>   district,
>   ward,
>   community_area,
>   latitude,
>   longitude,
>   crime_year
> FROM crime_raw;
Query ID = hadoop_20260509061541_5ce105d7-f0bf-44b2-aef1-c33fd0e62533
Total jobs = 1
Launching Job 1 out of 1
Tez session was closed. Reopening...
Session re-established.
Status: Running (Executing on YARN cluster with App id application_1778305578809_0002)

-----
VERTICES    MODE      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 ..... container  SUCCEEDED  2      2      0      0      0      0
Reducer 2 ..... container  SUCCEEDED  1      1      0      0      0      0
-----
VERTICES: 02/02 [=====] 100% ELAPSED TIME: 29.63 s

[ { _id: 'delivered', order_count: 3 } ]
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce>
```

```
Terminal Shell Edit View Window Help
shreyash@hadoop[ip-172-31-61-105:~] -- ssh -i ~/Desktop/Big_Data/ec2-1.pem hadoop@ec2-54-236-51-218.compute-1.amazonaws.com

> block,
> primary_type,
> description,
> location_desc,
> arrest,
> domestic,
> beat,
> district,
> ward,
> community_area,
> latitude,
> longitude,
> crime_year
FROM crime_raw;
Query ID = hadoop_20260509061541_5ce105d7-f0bf-44b2-aef1-c33fd0e62533
Total jobs = 1
Launching Job 1 out of 1
Tez session was closed. Reopening...
Session re-established.
Status: Running (Executing on YARN cluster with App id application_1778305578809_0002)

-----
VERTICES      MODE      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
Map 1 ..... container  SUCCEEDED  2      2      0      0      0      0
Reducer 2 ..... container  SUCCEEDED  1      1      0      0      0      0
-----
VERTICES: 02/02 [=====] 100% ELAPSED TIME: 29.63 s
-----
Loading data to table default.crimes_partitioned partition (crime_year=null)

Loaded : 1/3 partitions.
Failed with exception Exception when loading 3 in table crimes_partitioned with loadPath=hdfs://ip-172-31-61-105.ec2.internal:8020/user/hive/warehouse/crimes_partitioned/.hive-staging_hive_2026-05-09_06-15-41_657_4126690345264643663-1/-ext-10000
FAILED: Execution Error, return code 1 from org.apache.hadoop.hive.q1.exec.MoveTask. Exception when loading 3 in table crimes_partitioned with loadPath=hdfs://ip-172-31-61-105.ec2.internal:8020/user/hive/warehouse
Loaded : 2/3 partitions.
hive>

[ { _id: 'delivered', order_count: 3 } ]
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce>
```

```
Terminal Shell Edit View Window Help
shreyash@hadoop[ip-172-31-61-105:~] -- ssh -i ~/Desktop/Big_Data/ec2-1.pem hadoop@ec2-54-236-51-218.compute-1.amazonaws.com

SELECT COUNT(*) AS total_crimes
FROM crimes_partitioned;
OK
0
Time taken: 0.585 seconds, Fetched: 1 row(s)
hive> SELECT COUNT(*) AS total_crimes
FROM crimes_partitioned;
OK
0
Time taken: 0.239 seconds, Fetched: 1 row(s)
hive> SELECT
> primary_type,
> COUNT(*) AS crime_count
> FROM crimes_partitioned
> GROUP BY primary_type
> ORDER BY crime_count DESC
> LIMIT 5;
Query ID = hadoop_20260509061831_d61fb3bd-edda-44e9-a8e7-c24506092251
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1778305578809_0002)

-----
VERTICES      MODE      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
Map 1 ..... container  SUCCEEDED  1      1      0      0      0      0
Reducer 2 ..... container  SUCCEEDED  1      1      0      0      0      0
Reducer 3 ..... container  SUCCEEDED  1      1      0      0      0      0
-----
VERTICES: 03/03 [=====] 100% ELAPSED TIME: 8.89 s
-----
OK
NULL      232934
THEFT      54900
BATTERY    40970
CRIMINAL DAMAGE 27250
MOTOR VEHICLE THEFT      21473
Time taken: 9.789 seconds, Fetched: 5 row(s)
hive>

[ { _id: 'delivered', order_count: 3 } ]
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce>
```

```
Terminal Shell Edit View Window Help Sat 9 May 11:49 AM

shreyash — hadoop@ip-172-31-61-105:~ -- ssh -i ~/Desktop/Big_Data/ec2-1.pem hadoop@ec2-54-236-51-218.compute-1.amazonaws.com

Map 1 ..... container SUCCEEDED 1 1 0 0 0 0
Reducer 2 ..... container SUCCEEDED 1 1 0 0 0 0
Reducer 3 ..... container SUCCEEDED 1 1 0 0 0 0

VERTICES: 03/03 [=====] 100% ELAPSED TIME: 8.89 s

OK
NULL 232934
THEFT 54900
BATTERY 40970
CRIMINAL DAMAGE 27250
MOTOR VEHICLE THEFT 21473
Time taken: 9.789 seconds, Fetched: 5 row(s)
hive> SELECT
> crime_year,
> COUNT(*) AS total_crimes
> FROM crimes_partitioned
> GROUP BY crime_year
> ORDER BY crime_year ASC;
Query ID = hadoop_20260509061911_0b912da2-72e4-4287-aa9c-09a9ffbd2a78
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1778305578809_0002)

VERTICES MODE STATUS TOTAL COMPLETED RUNNING PENDING FAILED KILLED
Map 1 ..... container SUCCEEDED 1 1 0 0 0 0
Reducer 2 ..... container SUCCEEDED 1 1 0 0 0 0
Reducer 3 ..... container SUCCEEDED 1 1 0 0 0 0

VERTICES: 03/03 [=====] 100% ELAPSED TIME: 6.75 s

OK
NULL 233412
2022 240065
Time taken: 7.508 seconds, Fetched: 2 row(s)
hive>
[ { _id: 'delivered', order_count: 3 } ]
Atlas atlas-8fh6nv-shard-0 [primary] ecommerce>
```

```
Terminal Shell Edit View Window Help Sat 9 May 11:50 AM

shreyash — hadoop@ip-172-31-61-105:~ -- ssh -i ~/Desktop/Big_Data/ec2-1.pem hadoop@ec2-54-236-51-218.compute-1.amazonaws.com

hive> SELECT
> primary_type,
> COUNT(*) AS arrest_count
> FROM crimes_partitioned
> WHERE arrest = true
> GROUP BY primary_type
> HAVING COUNT(*) > 500
> ORDER BY arrest_count DESC;
Query ID = hadoop_20260509061940_8b3ff522-399a-40b0-a867-74c2ca8d6ba3
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1778305578809_0002)

VERTICES MODE STATUS TOTAL COMPLETED RUNNING PENDING FAILED KILLED
Map 1 ..... container SUCCEEDED 1 1 0 0 0 0
Reducer 2 ..... container SUCCEEDED 1 1 0 0 0 0
Reducer 3 ..... container SUCCEEDED 1 1 0 0 0 0

VERTICES: 03/03 [=====] 100% ELAPSED TIME: 7.28 s

OK
THEFT 54900
BATTERY 40970
CRIMINAL DAMAGE 27250
MOTOR VEHICLE THEFT 21473
ASSAULT 20815
DECEPTIVE PRACTICE 17177
OTHER OFFENSE 14626
ROBBERY 8966
WEAPONS VIOLATION 8781
BURGLARY 7595
NARCOTICS 4757
CRIMINAL TRESPASS 4223
OFFENSE INVOLVING CHILDREN 1908
CRIMINAL SEXUAL ASSAULT 1612
SEX OFFENSE 1243
HOMICIDE 737
PUBLIC PEACE VIOLATION 712
Time taken: 8.16 seconds, Fetched: 17 row(s)
```

```
Terminal Shell Edit View Window Help Sat 9 May 11:51 AM
shreyash — hadoop@ip-172-31-61-105:~ -- ssh -i ~/Desktop/Big_Data/ec2-1.pem hadoop@ec2-54-236-51-218.compute-1.amazonaws.com

Status: Running (Executing on YARN cluster with App id application_1778305578809_0002)

-----
VERTICES      MODE      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 ..... container  SUCCEEDED  1      1      0      0      0      0
Reducer 2 ..... container  SUCCEEDED  1      1      0      0      0      0
Reducer 3 ..... container  SUCCEEDED  1      1      0      0      0      0
-----
VERTICES: 03/03 [=====] 100% ELAPSED TIME: 7.28 s
-----

OK
THEFT 54900
BATTERY 40970
CRIMINAL DAMAGE 27250
MOTOR VEHICLE THEFT 21473
ASSAULT 20815
DECEPTIVE PRACTICE 17177
OTHER OFFENSE 14626
ROBBERY 8966
WEAPONS VIOLATION 8781
BURGLARY 7595
NARCOTICS 4757
CRIMINAL TRESPASS 4223
OFFENSE INVOLVING CHILDREN 1908
CRIMINAL SEXUAL ASSAULT 1612
SEX OFFENSE 1243
HOMICIDE 737
PUBLIC PEACE VIOLATION 712
Time taken: 0.16 seconds, Fetched: 17 row(s)
hive> SELECT *
> FROM crimes_partitioned
> WHERE crime_year = 2020;
OK
Time taken: 0.675 seconds
hive> SELECT *
> FROM crimes_partitioned
> WHERE crime_year = 2020;
OK
Time taken: 0.147 seconds
hive>
```

```
Terminal Shell Edit View Window Help Sat 9 May 11:51 AM
shreyash — hadoop@ip-172-31-61-105:~ -- ssh -i ~/Desktop/Big_Data/ec2-1.pem hadoop@ec2-54-236-51-218.compute-1.amazonaws.com

HOMICIDE 737
PUBLIC PEACE VIOLATION 712
Time taken: 0.16 seconds, Fetched: 17 row(s)
hive> SELECT *
> FROM crimes_partitioned
> WHERE crime_year = 2020;
OK
Time taken: 0.675 seconds
hive> SELECT *
> FROM crimes_partitioned
> WHERE crime_year = 2020;
OK
Time taken: 0.147 seconds
hive> SELECT
> district,
> COUNT(*) AS domestic_crime_count
> FROM crimes_partitioned
> WHERE domestic = true
> GROUP BY district
> ORDER BY domestic_crime_count DESC
> LIMIT 3;
Query ID = hadoop_20260509062124_cd535011-6f4a-4917-9f9f-0f96091a2b43
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1778305578809_0002)

-----
VERTICES      MODE      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 ..... container  SUCCEEDED  1      1      0      0      0      0
Reducer 2 ..... container  SUCCEEDED  1      1      0      0      0      0
Reducer 3 ..... container  SUCCEEDED  1      1      0      0      0      0
-----
VERTICES: 03/03 [=====] 100% ELAPSED TIME: 8.70 s
-----

OK
008 14873
006 14761
012 14415
Time taken: 0.362 seconds, Fetched: 3 row(s)
hive>
```

The screenshot displays the Databricks web interface. The top navigation bar includes links for File, Edit, View, History, Bookmarks, Profiles, Tab, Window, and Help. The browser address bar shows the URL: `dbc-874586be-6953.cloud.databricks.com/explore/data/volumes/vita_prac/vita_sche/vita_vol?o=7474646961268073`. The Databricks logo is in the top left corner, and a search bar is in the top right.

The left sidebar contains a navigation menu with the following items: New, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover (Beta), Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Spaces, Alerts, Query History, and SQL Warehouses. The 'Catalog' section is currently selected.

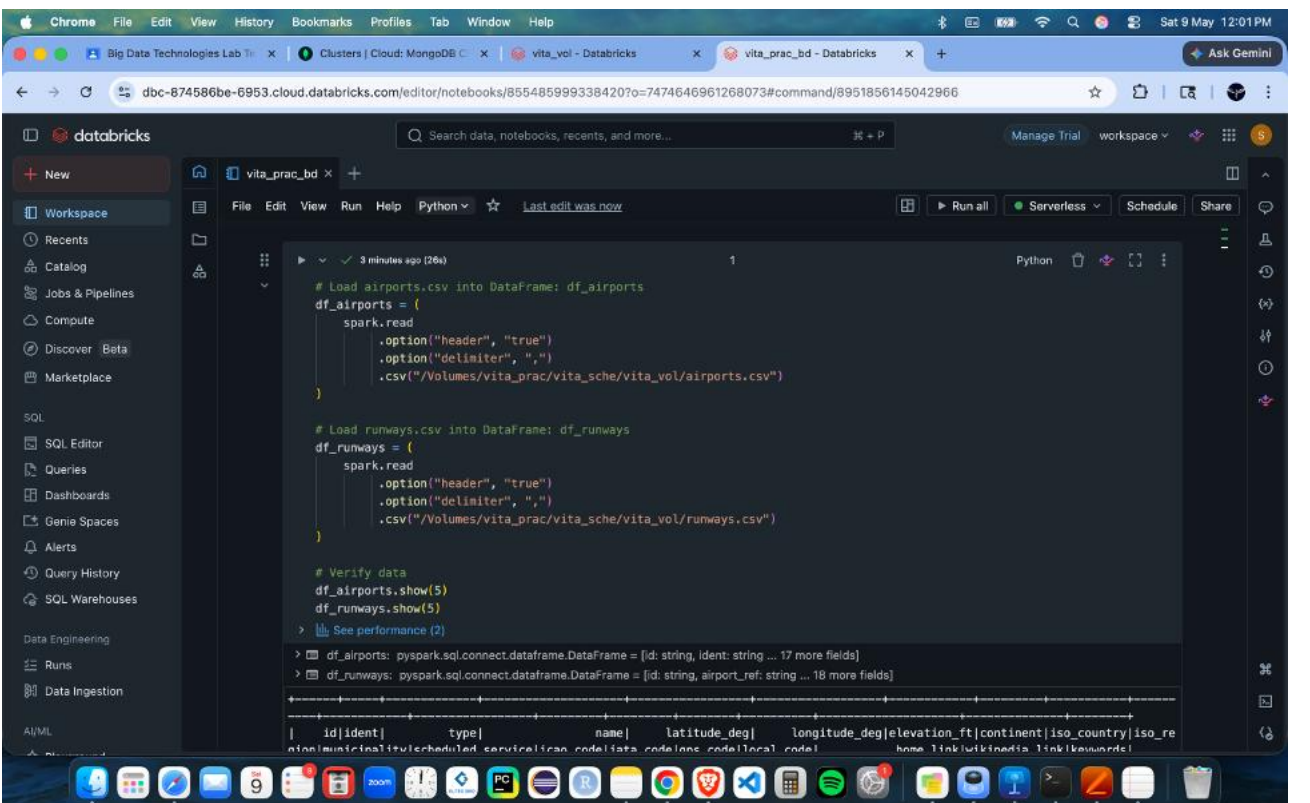
The main content area is divided into two panels. The left panel, titled 'Catalog', shows a tree view of the catalog structure. The right panel, titled 'vita_vol', shows the details of the selected volume. The 'Overview' tab is active, displaying a description, a table of files, and an 'About this volume' section.

The 'vita_vol' volume details include:

- Description:** A section with an 'AI generate' button and an 'Add' button.
- Files:** A table listing the files in the volume.

Name
airports.csv
runways.csv
- About this volume:** A section showing the owner as `ssdon2002@gmail.com` and an 'Add tags' button.

A toast notification at the bottom right indicates that 2 files were uploaded successfully. The notification text is: 'Upload summary: 2 files uploaded'. The files listed are `/Volumes/vita_prac/...ita_vol/airports.csv` and `/Volumes/vita_prac/...ita_vol/runways.csv`.



Chrome File Edit View History Bookmarks Profiles Tab Window Help

Big Data Technologies Lab T... Clusters | Cloud: MongoDB X vita_vol - Databricks X vita_prac_bd - Databricks X Ask Gemini

dbc-874586be-6953.cloud.databricks.com/editor/notebooks/85548599338420?o=7474646961268073#command/8951856145042966

databricks Search data, notebooks, recents, and more... Manage Trial workspace

New Workspace Recents Catalog Jobs & Pipelines Compute Discover Beta Marketplace

SQL SQL Editor Queries Dashboards Genie Spaces Alerts Query History SQL Warehouses

Data Engineering Runs Data Ingestion

AI/ML

File Edit View Run Help Python ☆ Last edit was 1 minute ago

4 minutes ago (26s) 1 Python

See performance (2)

df_airports: pyspark.sql.connect.dataframe.DataFrame = [id: string, ident: string ... 17 more fields]
df_runways: pyspark.sql.connect.dataframe.DataFrame = [id: string, airport_ref: string ... 18 more fields]

id	ident	type	name	latitude_deg	longitude_deg	elevation_ft	continent	iso_country	iso_re
6523	00A	heliport	Total RF Heliport	48.070985	-74.933689	11	NA	US	U
S-PA	Bensalem	no	NULL	NULL	00A	00A	https://www.pennd...	NULL	NULL
323361	00AA	small_airport	Aero B Ranch Airport	38.704022	-101.473911	3435	NA	US	U
S-KS	Leoti	no	NULL	NULL	00AA	00AA	NULL	NULL	NULL
6524	00AK	small_airport	Lowell Field	59.947733	-151.692524	450	NA	US	U
S-AK	Anchor Point	no	NULL	NULL	00AK	00AK	NULL	NULL	NULL
6525	00AL	small_airport	Epps Airpark	34.86479949951172	-86.77030181884766	820	NA	US	U
S-AL	Harvest	no	NULL	NULL	00AL	00AL	NULL	NULL	NULL
506791	00AN	small_airport	Katmai Lodge Airport	59.003287	-156.456699	80	NA	US	U
S-AK	King Salmon	no	NULL	NULL	00AN	00AN	NULL	NULL	NULL

only showing top 5 rows

1 minute ago (<1s) 2

```
from pyspark.sql.functions import col
```

Chrome File Edit View History Bookmarks Profiles Tab Window Help

Big Data Technologies Lab T... Clusters | Cloud: MongoDB X vita_vol - Databricks X vita_prac_bd - Databricks X Ask Gemini

dbc-874586be-6953.cloud.databricks.com/editor/notebooks/85548599338420?o=7474646961268073#command/8951856145042966

databricks Search data, notebooks, recents, and more... Manage Trial workspace

New Workspace Recents Catalog Jobs & Pipelines Compute Discover Beta Marketplace

SQL SQL Editor Queries Dashboards Genie Spaces Alerts Query History SQL Warehouses

Data Engineering Runs Data Ingestion

AI/ML

File Edit View Run Help Python ☆ Last edit was 1 minute ago

1 minute ago (<1s) 2 Python

```
from pyspark.sql.functions import col

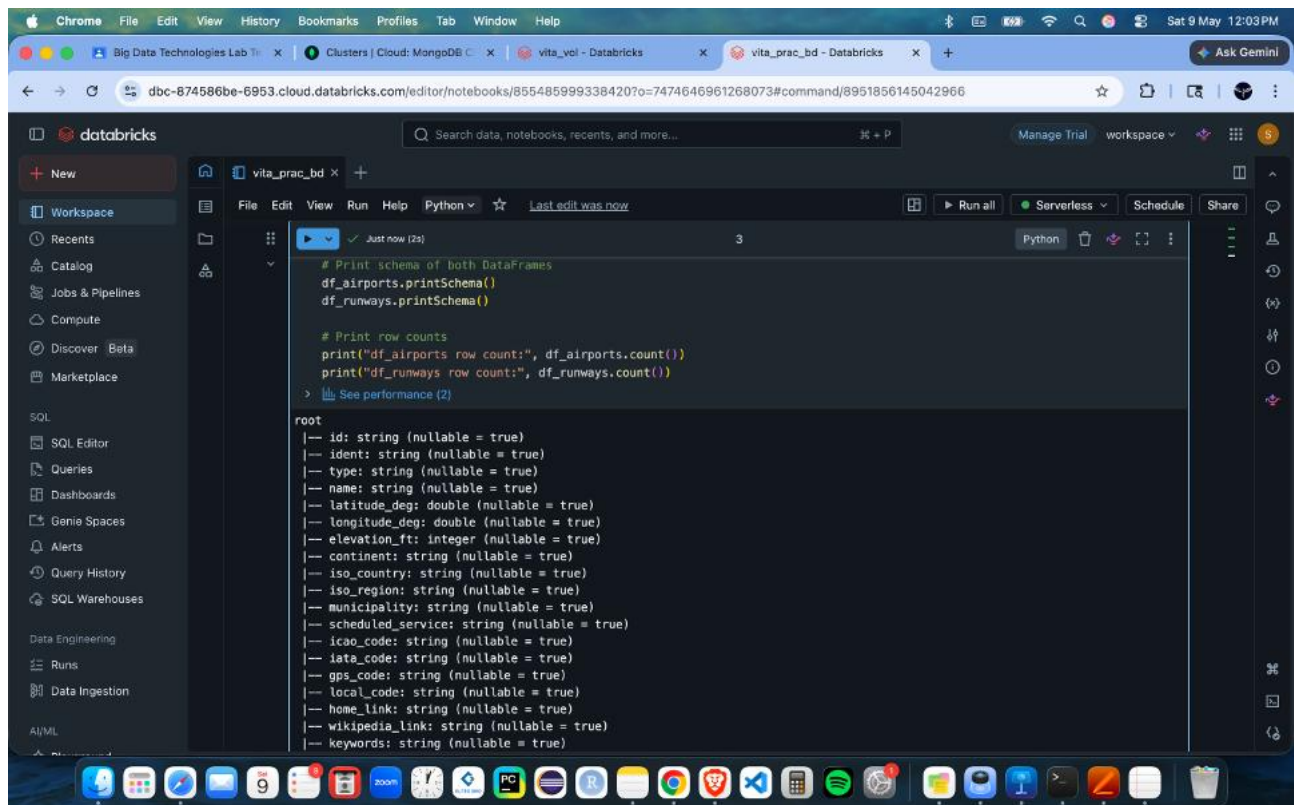
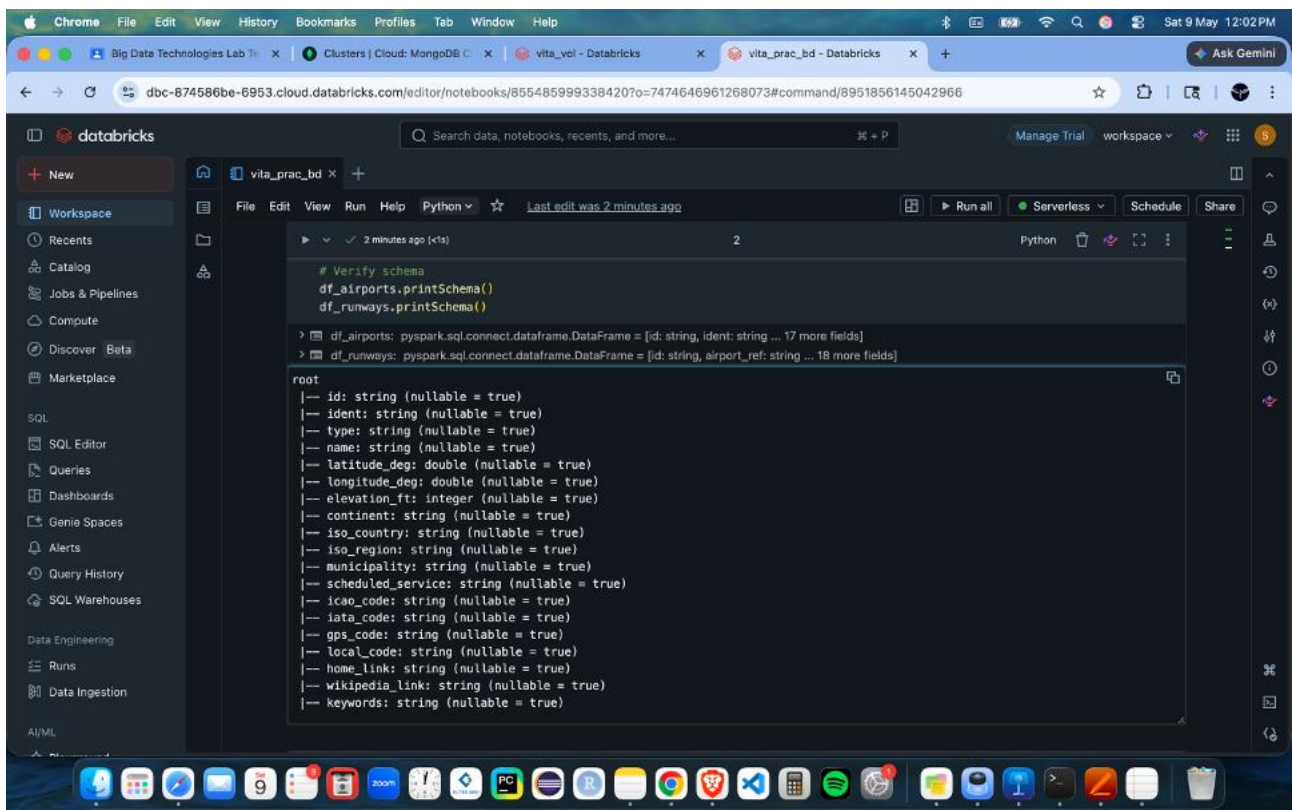
# Cast numerical columns in df_airports
df_airports = (
    df_airports
    .withColumn("latitude_deg", col("latitude_deg").cast("double"))
    .withColumn("longitude_deg", col("longitude_deg").cast("double"))
    .withColumn("elevation_ft", col("elevation_ft").cast("int"))
)

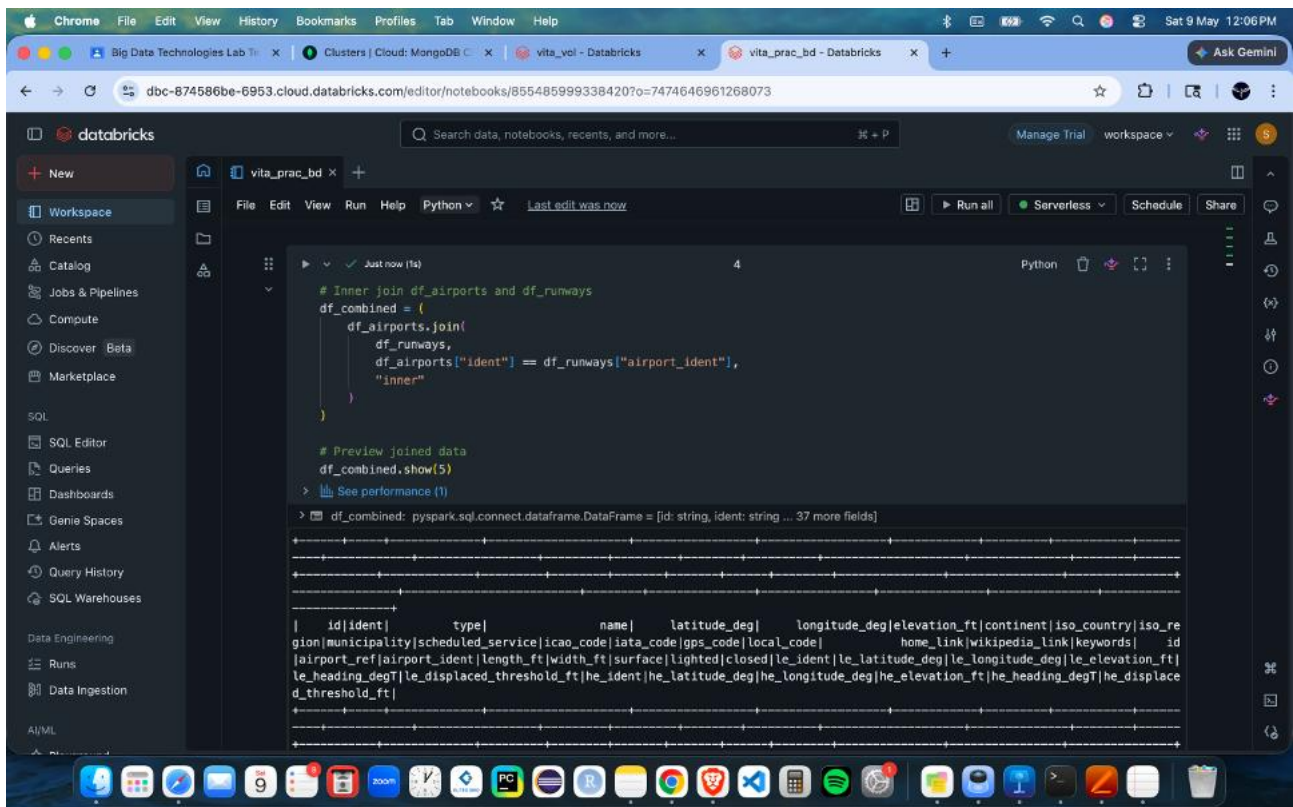
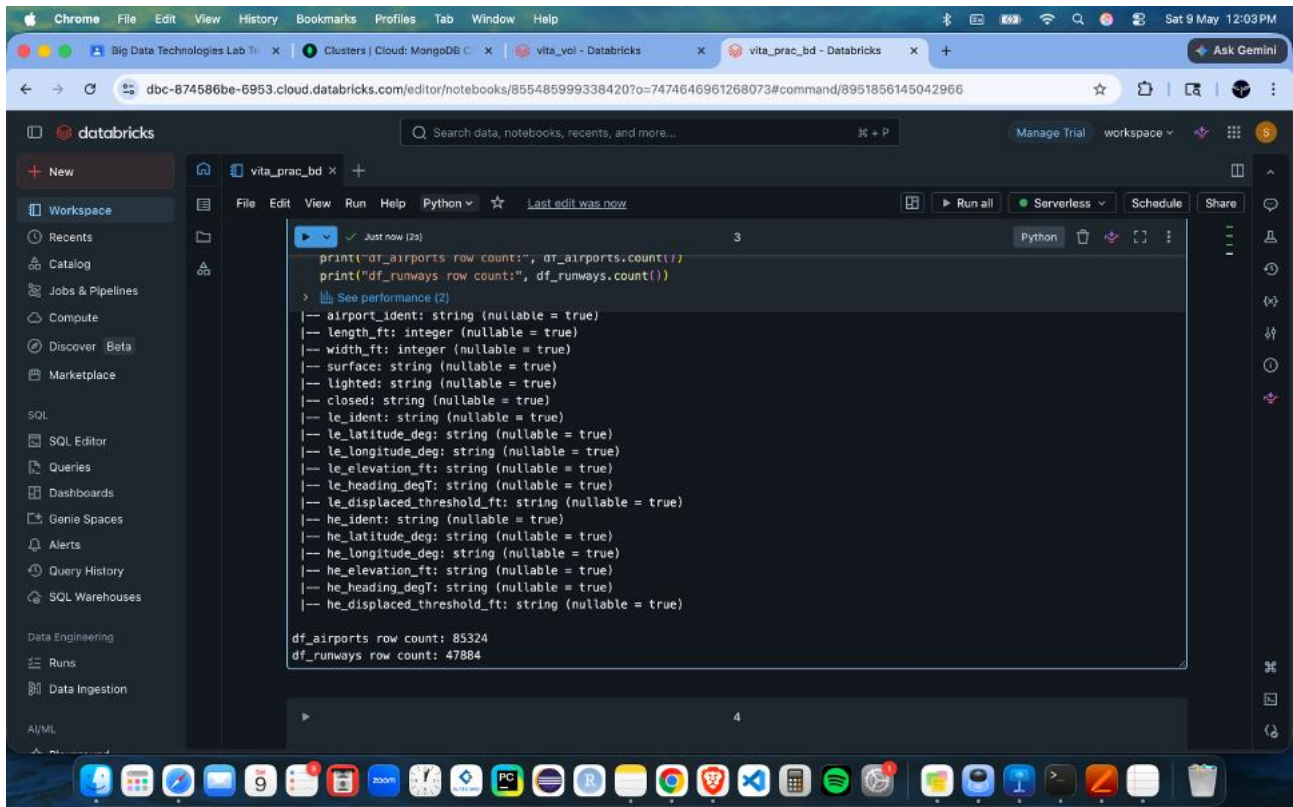
# Cast numerical columns in df_runways
df_runways = (
    df_runways
    .withColumn("length_ft", col("length_ft").cast("int"))
    .withColumn("width_ft", col("width_ft").cast("int"))
)

# Verify schema
df_airports.printSchema()
df_runways.printSchema()
```

df_airports: pyspark.sql.connect.dataframe.DataFrame = [id: string, ident: string ... 17 more fields]
df_runways: pyspark.sql.connect.dataframe.DataFrame = [id: string, airport_ref: string ... 18 more fields]

```
root
|-- id: string (nullable = true)
|-- ident: string (nullable = true)
|-- type: string (nullable = true)
```





Chrome File Edit View History Bookmarks Profiles Tab Window Help

Big Data Technologies Lab | Clusters | Cloud: MongoDB | vita_vol - Databricks | vita_prac_bd - Databricks

dbc-874586be-6953.cloud.databricks.com/editor/notebooks/85548599338420?o=7474646961268073

Search data, notebooks, recents, and more... Manage Trial workspace

New Workspace Recents Catalog Jobs & Pipelines Compute Discover Beta Marketplace SQL SQL Editor Queries Dashboards Genie Spaces Alerts Query History SQL Warehouses Data Engineering Runs Data Ingestion AI/ML

vita_prac_bd

File Edit View Run Help Python Last edit was now

Run all Serverless Schedule Share

Just now (1s) 5 Python

```
from pyspark.sql.functions import when, col

# Add runway_category column
df_combined = (
    df_combined
    .withColumn(
        "runway_category",
        when(col("length_ft") >= 10000, "Long")
        .when((col("length_ft") >= 5000) & (col("length_ft") < 10000), "Medium")
        .otherwise("Short")
    )
)

# Preview result
df_combined.select("id", "airport_id", "length_ft", "runway_category").show(10)
```

df_combined: pyspark.sql.connect.dataframe.DataFrame = [id: string, id: string ... 38 more fields]

id	airport_id	length_ft	runway_category
00A	00A	80	Short
00AK	00AK	2500	Short
00AL	00AL	2100	Short
00AN	00AN	4517	Short
00AS	00AS	1450	Short
00AZ	00AZ	1700	Short
00CA	00CA	6000	Medium

Chrome File Edit View History Bookmarks Profiles Tab Window Help

Big Data Technologies Lab | Clusters | Cloud: MongoDB | vita_vol - Databricks | vita_prac_bd - Databricks

dbc-874586be-6953.cloud.databricks.com/editor/notebooks/85548599338420?o=7474646961268073

Search data, notebooks, recents, and more... Manage Trial workspace

New Workspace Recents Catalog Jobs & Pipelines Compute Discover Beta Marketplace SQL SQL Editor Queries Dashboards Genie Spaces Alerts Query History SQL Warehouses Data Engineering Runs Data Ingestion AI/ML

vita_prac_bd

File Edit View Run Help Python Last edit was now

Run all Serverless Schedule Share

Just now (2s) 6 Python

```
# Create temporary view
df_combined.createOrReplaceTempView("airport_view")

spark.sql("""
SELECT
    iso_country,
    COUNT(*) AS runway_count
FROM airport_view
GROUP BY iso_country
ORDER BY runway_count DESC
LIMIT 10
""").show()
```

iso_country	runway_count
US	26450
BR	5825
AU	2103
CA	1578
AR	761
FR	659
GB	600
DE	592
RU	489
ID	414

Chrome File Edit View History Bookmarks Profiles Tab Window Help

Big Data Technologies Lab T... Clusters | Cloud: MongoDB C... vita_vol - Databricks vita_prac_bd - Databricks

dbc-874586be-6953.cloud.databricks.com/editor/notebooks/855485999338420?o=7474646961268073

Search data, notebooks, recent, and more... Manage Trial workspace

New Workspace Recents Catalog Jobs & Pipelines Compute Discover Beta Marketplace SQL SQL Editor Queries Dashboards Genie Spaces Alerts Query History SQL Warehouses Data Engineering Runs Data Ingestion AI/ML

File Edit View Run Help Python Last edit was now Run all Serverless Schedule Share

```
spark.sql("""
SELECT
    continent,
    AVG(length_ft) AS avg_runway_length
FROM airport_view
WHERE length_ft IS NOT NULL
GROUP BY continent
ORDER BY avg_runway_length DESC
""").show()
```

See performance (1)

continent	avg_runway_length
AN	7768.857142857143
AS	6925.6469996647675
AF	6787.526448362721
EU	4183.151335311572
OC	3558.0401785714284
SA	2857.0091141239963
NA	2599.724218941758

Mac OS X dock with various application icons.

Chrome File Edit View History Bookmarks Profiles Tab Window Help

Big Data Technologies Lab T... Clusters | Cloud: MongoDB C... vita_vol - Databricks vita_prac_bd - Databricks

dbc-874586be-6953.cloud.databricks.com/editor/notebooks/855485999338420?o=7474646961268073#command/4734958004600758

Search data, notebooks, recent, and more... Manage Trial workspace

New Workspace Recents Catalog Jobs & Pipelines Compute Discover Beta Marketplace SQL SQL Editor Queries Dashboards Genie Spaces Alerts Query History SQL Warehouses Data Engineering Runs Data Ingestion AI/ML

File Edit View Run Help Python Last edit was now Run all Serverless Schedule Share

```
spark.sql("""
SELECT
    name,
    municipality,
    COUNT(*) AS lighted_runway_count
FROM airport_view
WHERE iso_country = 'IN'
AND lighted = '1'
GROUP BY name, municipality
HAVING COUNT(*) >= 1
ORDER BY lighted_runway_count DESC
""").show()
```

See performance (1)

name	municipality	lighted_runway_count
[Indira Gandhi Int...]	New Delhi	3
[Navi Mumbai Inter...]	Navi Mumbai	2
[Raja Bhoj Interna...]	Bhopal	2
[Rajiv Gandhi Inte...]	Hyderabad	2
[Netaji Subhash Ch...]	Kolkata	2
[Agra Airport / Ag...]	Agra	2
[Prayagraj Airport]	Allahabad	2
[Ambala Air Force ...]	Ambala	2
[Chennai Internati...]	Chennai	2
[Kempegowda Intern...]	Bengaluru	2
[Mangaluru Interna...]	Mangaluru	2

Mac OS X dock with various application icons.

Chrome File Edit View History Bookmarks Profiles Tab Window Help

Big Data Technologies Lab Th x Clusters | Cloud: MongoDB C x vita_vol - Databricks x vita_prac_bd - Databricks x Ask Gemini

dbc-874586be-6953.cloud.databricks.com/editor/notebooks/855485999338420?o=7474646961268073#command/4734958004600757

databricks Search data, notebooks, recents, and more... Manage Trial workspace

New Workspace Recents Catalog Jobs & Pipelines Compute Discover Beta Marketplace SQL SQL Editor Queries Dashboards Genie Spaces Alerts Query History SQL Warehouses Data Engineering Runs Data Ingestion AI/ML

vita_prac_bd x

File Edit View Run Help Python Last edit was now Run all Serverless Schedule Share

Just now (1s) 8 Python

```
ORDER BY lighted_runway_count DESC
""").show()
> See performance (1)
```

[Navi Mumbai Inter...]	Navi Mumbai	2
[Raja Bhoj Interna...]	Bhopal	2
[Rajiv Gandhi Inte...]	Hyderabad	2
[Netaji Subhash Ch...]	Kolkata	2
[Agra Airport / Ag...]	Agra	2
[Prayagraj Airport]	Allahabad	2
[Ambala Air Force ...]	Ambala	2
[Chennai Internati...]	Chennai	2
[Kempegowda Intern...]	Bengaluru	2
[Mangaluru Interna...]	Mangaluru	2
[Bidar Airport / B...]	Bidar	2
[Chhatrapati Shiva...]	Mumbai	2
[Tambaram Air Forc...]	Chennai	2
[Jammu Airport]	Jammu	1
[Dhulia Airport]	NULL	1
[Shravasti Airport]	Shravasti	1
[Raigarh Airport (...]	NULL	1
[Maharshi Valmiki ...]	Faizabad	1
[Jodhpur Airport]	Jodhpur	1

only showing top 20 rows

9

Chrome File Edit View History Bookmarks Profiles Tab Window Help

Big Data Technologies Lab Th x Clusters | Cloud: MongoDB C x vita_vol - Databricks x vita_prac_bd - Databricks x Ask Gemini

dbc-874586be-6953.cloud.databricks.com/editor/notebooks/855485999338420?o=7474646961268073

databricks Search data, notebooks, recents, and more... Manage Trial workspace

New Workspace Recents Catalog Jobs & Pipelines Compute Discover Beta Marketplace SQL SQL Editor Queries Dashboards Genie Spaces Alerts Query History SQL Warehouses Data Engineering Runs Data Ingestion AI/ML

vita_prac_bd x

File Edit View Run Help Python Last edit was now Run all Serverless Schedule Share

Just now (1s) 9 Python

```
spark.sql("""
SELECT
    name,
    iso_country,
    MAX(length_ft) AS max_runway_length
FROM airport_view
GROUP BY name, iso_country
ORDER BY max_runway_length DESC
LIMIT 5
""").show()
> See performance (1)
```

	name	iso_country	max_runway_length
[Gunflint Seaplane...]		US	30000
[Libby Camps Seapl...]		US	26000
[Brookville Reserv...]		US	25000
[Long Lake Seaplan...]		US	25000
[Conchas Lake Seap...]		US	21120

10

Start typing or generate with AI (* + I)...

